

ELECTRONIC DEVICES & CIRCUITS

SEMESTER PROJECT



**DEPARTMENT OF ELECTRICAL ENGINEERING
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ELECTRONIC DEVICES & CIRCUITS Semester Project Report

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1. Abstract:

DC power supplies are essential in electronics laboratories for testing and powering circuits. This project aims to design and build a dual-output power supply that includes one variable voltage and one fixed voltage output. The design utilizes rectifier circuits, filters, and voltage regulators to convert AC mains into stable DC voltages.

2. Project Requirements:

- Two independent DC outputs
- **Channel 1:** Adjustable 2–24V DC, 1A
- **Channel 2:** Fixed 5V DC, 1A
- AC to DC conversion
- Stable, low-ripple output
- Proper thermal and power dissipation handling

Apparatus:

- Step down Center Tapped Transformer (12-0-12V)
- Bridge Rectifier using Silicon diodes
- Capacitors
- Resistors
- 56ohm resistor
- BJT
- Potentiometer(5kohms)
- Voltage Regulators IC LM(317)

Tool used:

- Proteus

3.DC Power Supply

Introduction:

A dual output power supply is a type of power supply that can provide two separate and independent DC output voltages. Most DC power supplies have two modes of operation. In Constant Voltage (CV) mode, the power supply controls the output voltage based on the user settings. In Constant Current (CC) mode, the power supply regulates the current. These outputs can be used for various applications, such as powering multiple devices or circuits that require different voltage levels.

Working Principle:

The working principle of a dual output power supply is based on the use of a center-tapped transformer and a rectifier circuit. The center-tapped transformer is used to step down the input voltage to a lower level, while the rectifier circuit is used to convert the AC voltage into a DC voltage. The center-tapped transformer has an additional tap in the middle of the primary winding, which is connected to the center point of the secondary winding. This tap forms two separate outputs, with each output having a voltage equal to half of the total voltage of the secondary winding.

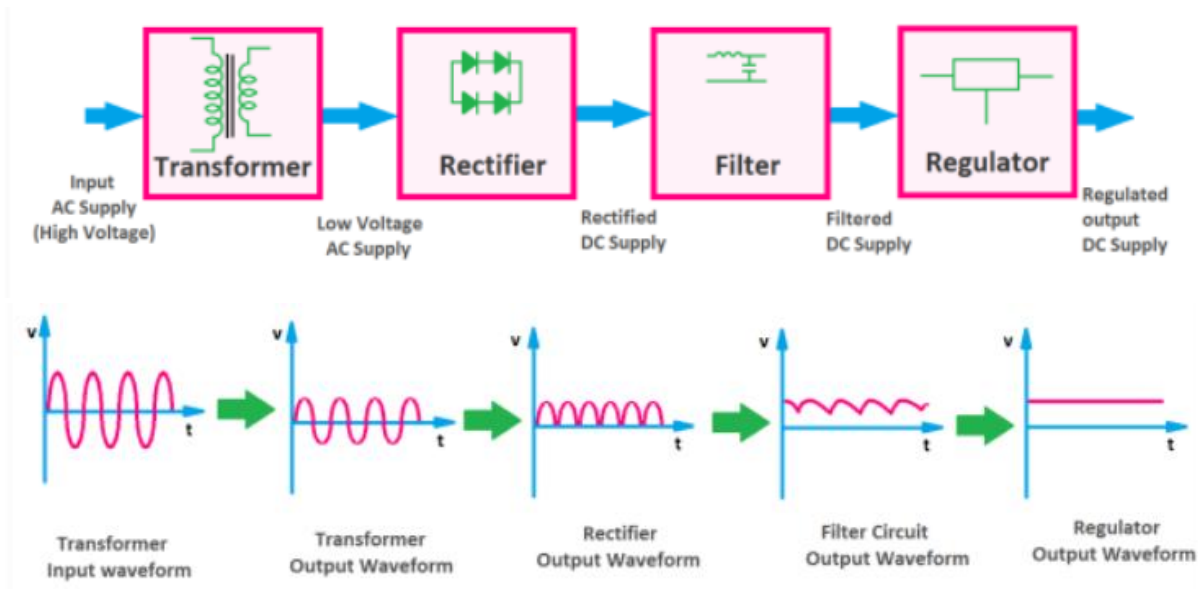
Rectifier:

The rectifier circuit is used to convert the AC voltage from the transformer into a DC voltage. A rectifier is made up of two diodes that are connected in a specific configuration. This configuration allows for the half-wave rectification of the AC voltage, which means that the current can only flow in one direction. The rectifier is connected to the secondary winding of the transformer

4. Circuit Topology & Block Diagram:

The power supply consists of the following stages:

1. Step-down Transformer
2. Full-wave Bridge Rectifier
3. Filter Capacitor
4. Voltage Regulator (LM317 for variable, 7805 for fixed)
5. Output Terminals & Load



The transformer steps down AC voltage; the rectifier converts it to pulsating DC; the capacitor smooths it; and the regulators provide stable DC at the required levels.

5. Calculations:

Ripple Voltage Calculation:

To size the input capacitors, we can use:

$$C = \frac{I}{2f \cdot V_r}$$

- I = Load current (A)

- f = Frequency after rectification (e.g., 100 Hz for 50 Hz input)
- V_{ripple} = Acceptable ripple voltage (V)

Where $I=1A$, $f=50\text{hz}$, $C= 1000 \text{ uF}$ (for stability)

Voltage Regulation Calculations for LM317

Keeping 56 Ohm fixed

The output voltage for the LM317 can be calculated with the formula:

$$V_{out} = 1.25 \left(1 + \frac{R_2}{R_1} \right)$$

For Minimum Voltage:

Given:

- $V_{out} = 2V$
- $R_1 = 56 \Omega$

$$V_{out} = 1.25 \left(1 + \frac{R_2}{R_1} \right)$$

Calculating R_2 :

$$R_2 = 33 \Omega$$

For Maximum Voltage:

Given:

- $V_{out} = 24V$
- $R_1 = 56 \Omega$

Calculating R_2 :

$$R_2 = R_1 \left(\frac{V_{out}}{1.25} - 1 \right)$$

Substituting the values:

$$R_2 = 56 \left(\frac{24}{1.25} - 1 \right)$$

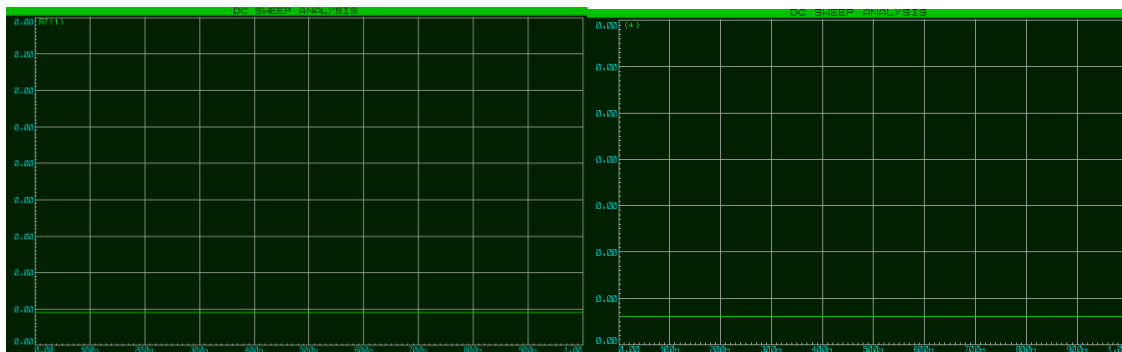
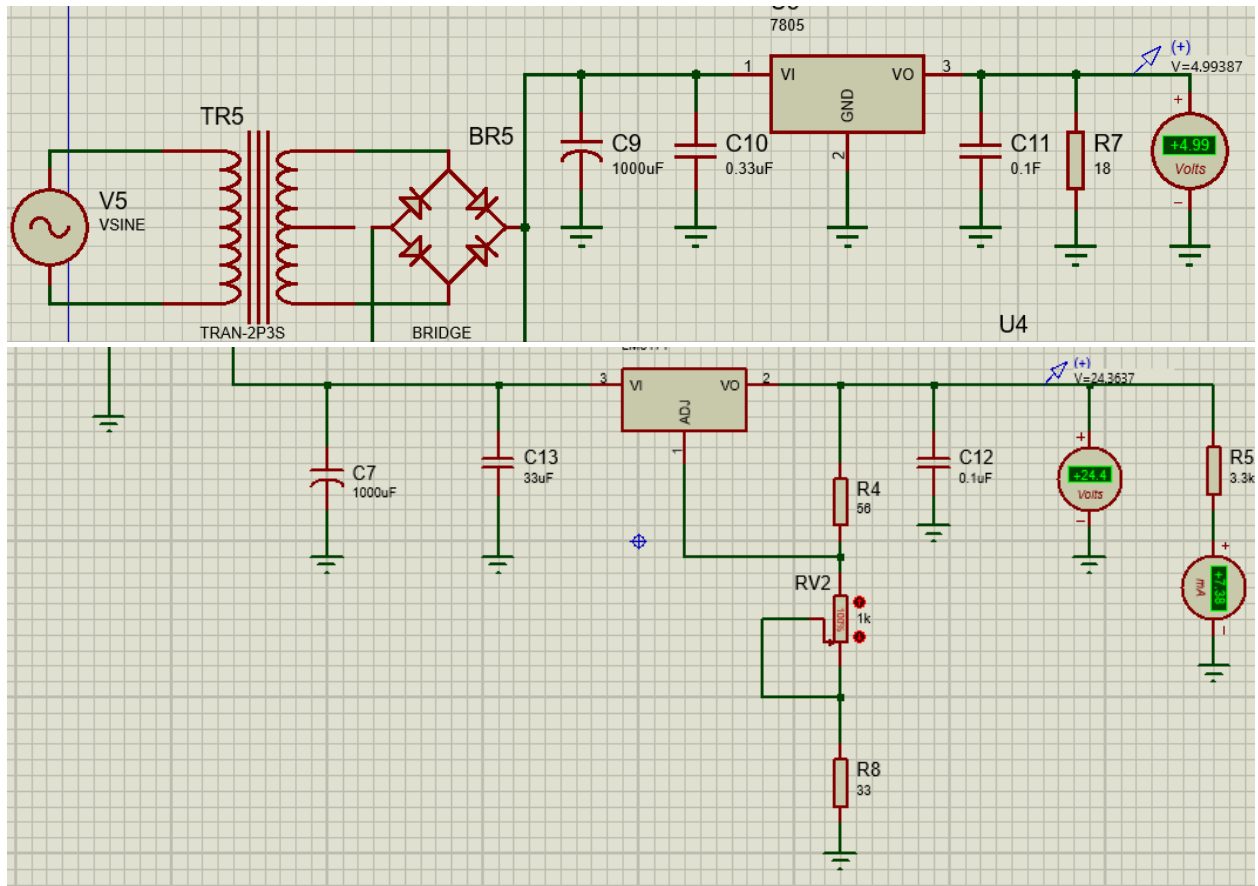
$$R_2 = 56(19.2-1) = 560 \times 18.2 = \mathbf{1011 \Omega}$$

Standard Value: Use a **1k ohm** variable resistor.

6. Simulation & Results:

Simulation was done in Proteus.

Schematic Diagram:



The observed results show:

- Smooth DC output after filtering
- LM317 output adjustable from 2–24V

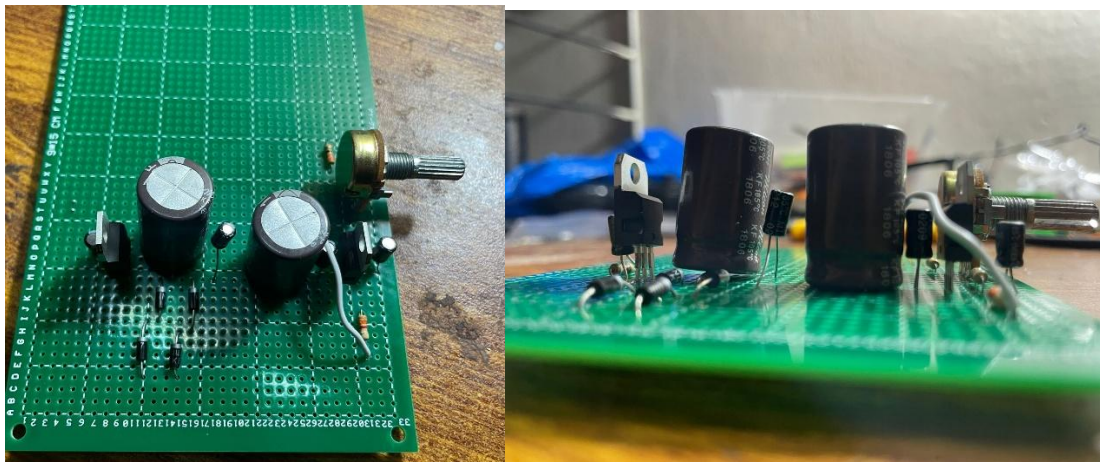
- 7805 giving constant 5V output

7. Hardware Implementation:

The circuit was first assembled on a breadboard.

- Rectifier and filter tested individually
- Regulator outputs checked with a multimeter
- Some variations in output were fixed using better grounding.

Final hardware produced stable regulated outputs.



8. Testing & Output Results

- **Channel 1:** Output varied smoothly from 2V to 24V under load



- **Channel 2:** Maintained constant 5V



9. Conclusion:

The designed power supply meets all required specifications. Both output channels deliver stable and regulated voltages with good performance in simulation and hardware. The project successfully demonstrates AC-to-DC conversion, regulation, and practical testing of electronic components.

10. References:

- LM317 and 7805 datasheets.
 - Lecture slides.
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